

Technology Stack

Date	04 July 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>Streamlit</i>	<i>To build an interactive and user-friendly web interface for voice analysis</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python 3.11</i>	<i>Handles application logic, AI processing, and report generation</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>Local File Storage / JSON Files</i>	<i>Stores audio files, concept data, and generated analysis reports</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Streamlit Cloud</i>	<i>Deploys the application online for easy user access</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Maintains source code, collaboration, and project version tracking</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>OpenAI Whisper, Sentence Transformers, Librosa, ReportLab</i>	<i>Enables speech-to-text conversion, concept similarity analysis, audio feature extraction, and PDF report generation</i>